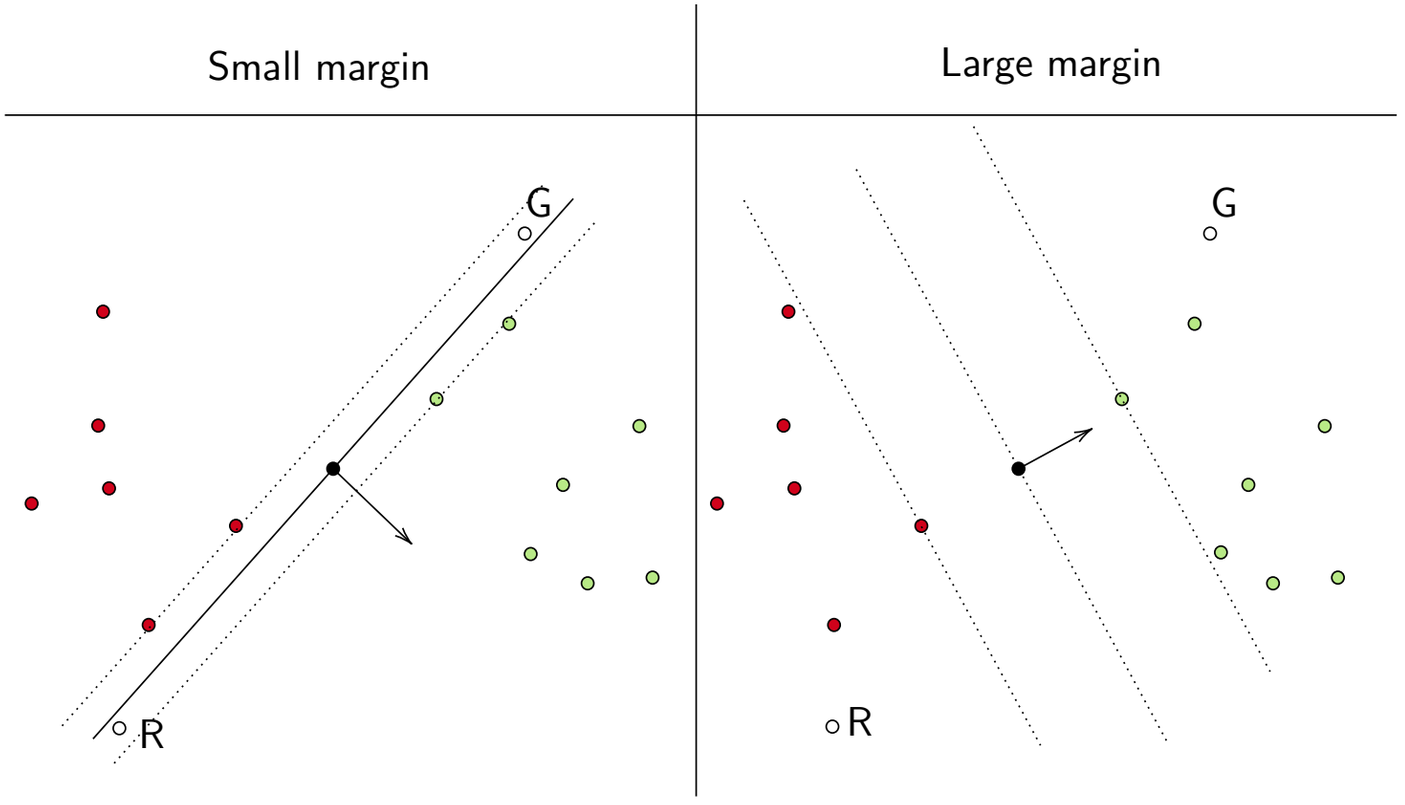
Week-10

Karthik Thiagarajan

1. Margin 2
2. Hard-Margin Support Vector Machines 4
3. Solving the Optimization Problem 6
   3.1. Primal 6
   3.2. Dual 7
   3.3. Solution 9
   3.4. Support Vectors 10
      3.4.1. Primal View 10
      3.4.2. Dual view 11
4. Prediction 13
5. Example 13
6. Kernel SVM 17

1. Margin

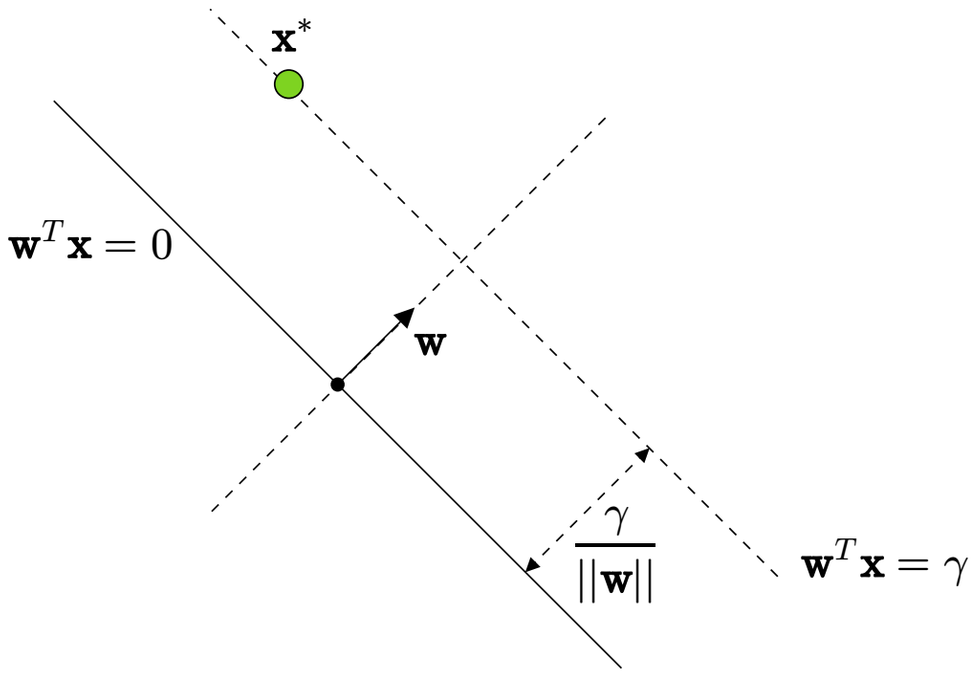
Classifiers with a larger margin generalize better. That is, they perform better on test data-points compared to classifiers that have a small margin. Here is a visual intuition for to justify this:



Given a linear classifier, let us make the notion of margin more precise. There are two kinds of margin. Consider a linearly separable dataset with a positive margin. Let $\mathbf{w}$ be any valid classifier that perfectly separates the dataset. We can always find at least one point from the dataset that is closest to the decision boundary. Call this point $\mathbf{x}^*$. For convenience, let this point have $y = 1$. This point will lie on some line parallel to the decision boundary, say $\mathbf{w}^T\mathbf{x} = \gamma$. This γ is termed the **functional margin**. From our assumption, we have:

$$\left(\mathbf{w}^T\mathbf{x}_i\right)y_i \geqslant \gamma,\ 1 \leqslant i \leqslant n$$

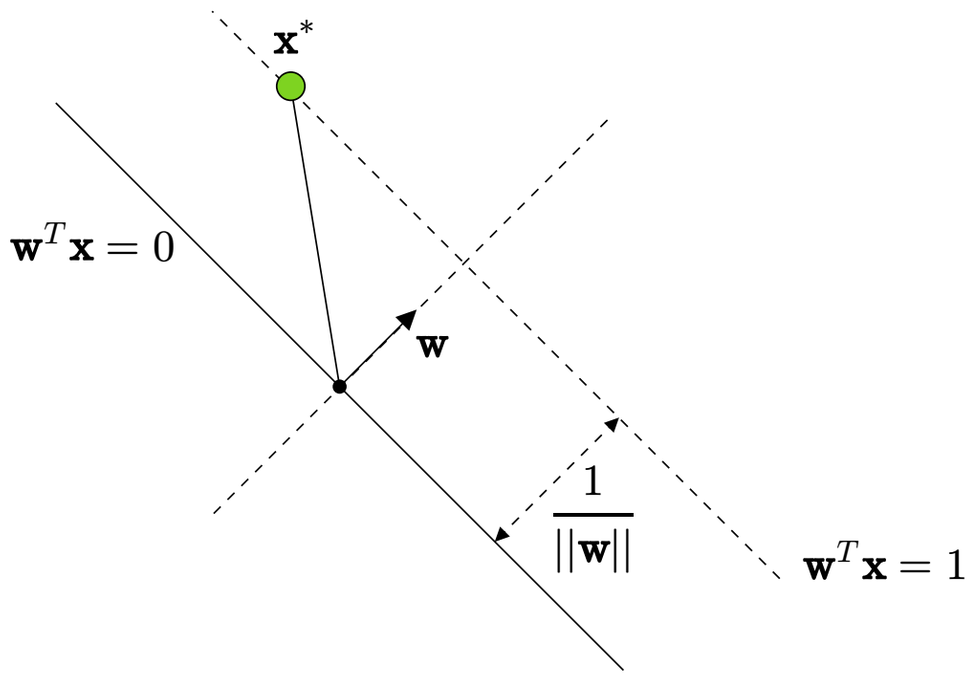
Visually:



The distance of between the lines $\mathbf{w}^T\mathbf{x} = 0$ and $\mathbf{w}^T\mathbf{x} = \gamma$ is called the geometric margin and is given by:

$$\frac{\mathbf{w}^T\mathbf{x}^*}{||\mathbf{w}||} = \frac{\gamma}{||\mathbf{w}||}$$

Since both $\mathbf{w}$ and γ can be scaled simultaneously without affecting the setup, we will set $\gamma = 1$. The corresponding figure becomes:



Therefore, for any linearly separable dataset with a positive margin, and a valid classifier $\mathbf{w}$, we can set the functional margin $\gamma = 1$ and the geometric margin becomes:

$$\text{(geometric) margin} \;=\; \frac{1}{||\mathbf{w}||}$$

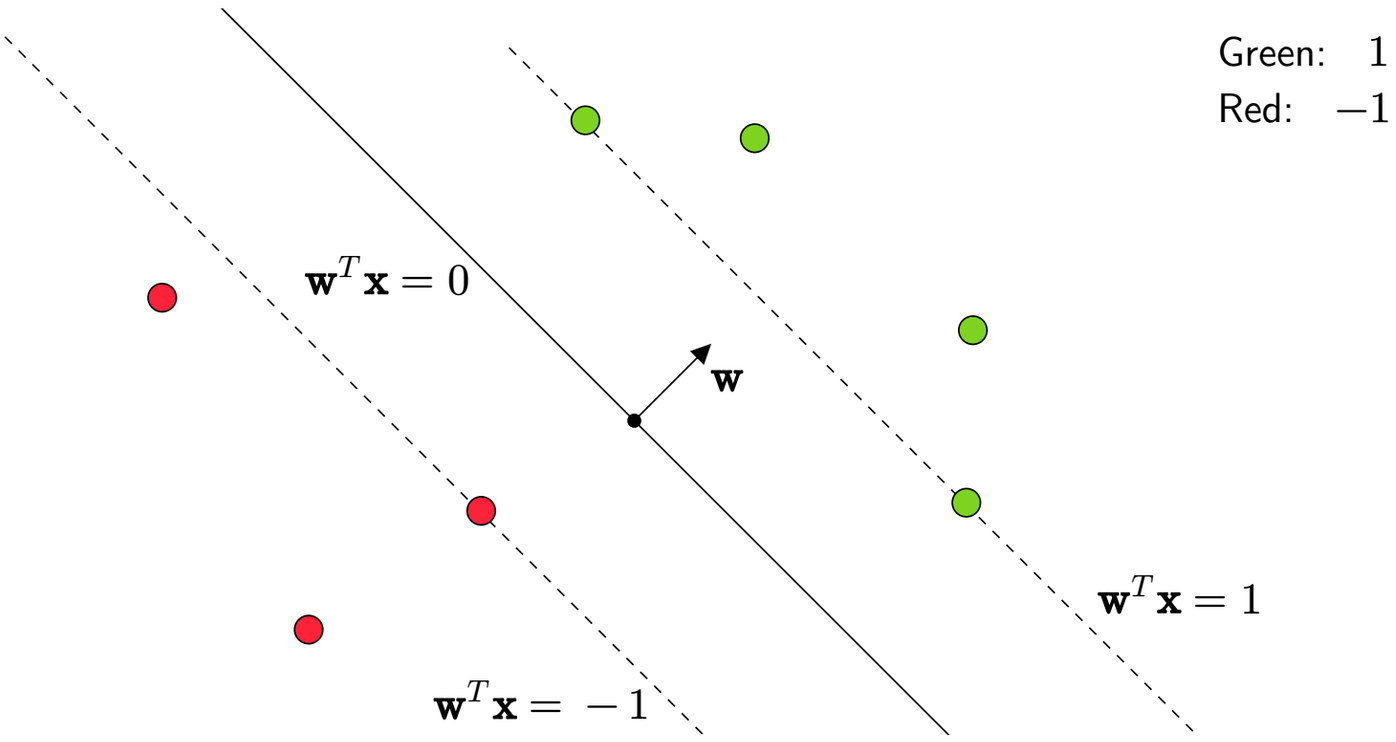
The margin is often used to denote both the distance as well as the hyperplanes $\mathbf{w}^T\mathbf{x} = \pm 1$. The context will determine what it means.

2. Hard-Margin Support Vector Machines

Now that the (geometric) margin is defined, we would like to find a classifier that has the maximum margin:

$$\max_{\mathbf{w}} \quad \frac{1}{||\mathbf{w}||}$$

But we also have constraints:



Since the closest points to the classifier lie on the margin, all other points have to lie beyond the margin, giving us the following n constraints:

$$\left(\mathbf{w}^T\mathbf{x}_i\right)y_i \geqslant 1,\ 1 \leqslant i \leqslant n$$

Putting these two together, we have:

$$\max_{\mathbf{w}} \ \frac{1}{||\mathbf{w}||}$$

subject to

$$\left(\mathbf{w}^T\mathbf{x}_i\right)y_i \geqslant 1,\ 1 \leqslant i \leqslant n$$

A slight modification to the objective function results in:

$$\min_{\mathbf{w}} \frac{||\mathbf{w}||^2}{2}$$

subject to

$$\left(\mathbf{w}^T\mathbf{x}_i\right)y_i \geqslant 1,\ 1 \leqslant i \leqslant n$$

3. Solving the Optimization Problem

3.1. Primal

Since the objective function is convex and since the inequality constraints involve convex functions, the entire problem is a convex optimization problem. We can recast it in the standard form as:

$$\min_{\mathbf{w}} \frac{||\mathbf{w}||^2}{2}$$

subject to

$$1 - \left(\mathbf{w}^T\mathbf{x}_i\right)y_i \leqslant 0,\ 1 \leqslant i \leqslant n$$

We will call this the primal form. In the primal form, the variable we are optimizing over is $\mathbf{w}$ and the constraints are called the primal constraints. There are n constraints, one corresponding to each data-point. We now introduce the

Lagrangian function with α_i being the ith Lagrange multiplier:

$$\frac{||\mathbf{w}||^2}{2} + \sum_{i=1}^{n} \alpha_i \Big[1 - \big(\mathbf{w}^T \mathbf{x}_i\big) y_i\Big]$$

We can equivalently express the original optimization problem as follows:

$$\min_{\mathbf{w}} \ \max_{\boldsymbol{\alpha} \geqslant \mathbf{0}} \ \frac{||\mathbf{w}||^2}{2} + \sum_{i=1}^{n} \alpha_i \Big[1 - \big(\mathbf{w}^T \mathbf{x}_i\big) y_i\Big]$$

3.2. Dual

The dual problem can be expressed as:

$$\max_{\boldsymbol{\alpha} \geqslant \mathbf{0}} \ \min_{\mathbf{w}} \ \frac{||\mathbf{w}||^2}{2} + \sum_{i=1}^{n} \alpha_i \Big[1 - \big(\mathbf{w}^T \mathbf{x}_i\big) y_i\Big]$$

The dual objective is itself an unconstrained optimization problem which we will solve now:

$$\min_{\mathbf{w}} \ \frac{||\mathbf{w}||^2}{2} + \sum_{i=1}^{n} \alpha_i \Big[1 - \big(\mathbf{w}^T \mathbf{x}_i\big) y_i\Big]$$

Setting the gradient to zero, we get the weight vector to be a linear combination of the data-points:

$$\mathbf{w} = \sum_{i=1}^{n} \alpha_i \mathbf{x}_i y_i$$

In matrix notation:

$$\mathbf{w} = \mathbf{X}\mathbf{Y}\boldsymbol{\alpha}$$

Here $\mathbf{X}$ is the $d \times n$ data-matrix, $\mathbf{Y}$ is a $n \times n$ diagonal matrix whose diagonal elements are made up of the labels of the n data-points. $\boldsymbol{\alpha}$ is a $n \times 1$ vector that has the Lagrange multipliers.

$$\mathbf{X} = \begin{bmatrix} | & & | \\ x_1 & \cdots & x_n \\ | & & | \end{bmatrix}, \ \mathbf{Y} = \begin{bmatrix} y_1 & & \\ & \ddots & \\ & & y_n \end{bmatrix}, \ \boldsymbol{\alpha} = \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{bmatrix}$$

Plugging $\mathbf{w} = \mathbf{X}\mathbf{Y}\boldsymbol{\alpha}$ back into the dual objective, we get the following form:

$$\sum_{i=1}^{n} \alpha_i - \frac{1}{2}\boldsymbol{\alpha}^T \left(\mathbf{Y}^T\mathbf{X}^T\mathbf{X}\mathbf{Y}\right)\boldsymbol{\alpha}$$

The dual optimization problem therefore becomes:

$$\boxed{\max_{\boldsymbol{\alpha} \geqslant \mathbf{0}} \quad \boldsymbol{\alpha}^T \mathbf{1} - \frac{1}{2}\boldsymbol{\alpha}^T \left(\mathbf{Y}^T\mathbf{X}^T\mathbf{X}\mathbf{Y}\right)\boldsymbol{\alpha}}$$

Here, $\mathbf{1}$ represents a vector of n ones. The objective function is a quadratic function of $\boldsymbol{\alpha}$ and is concave.

3.3. Solution

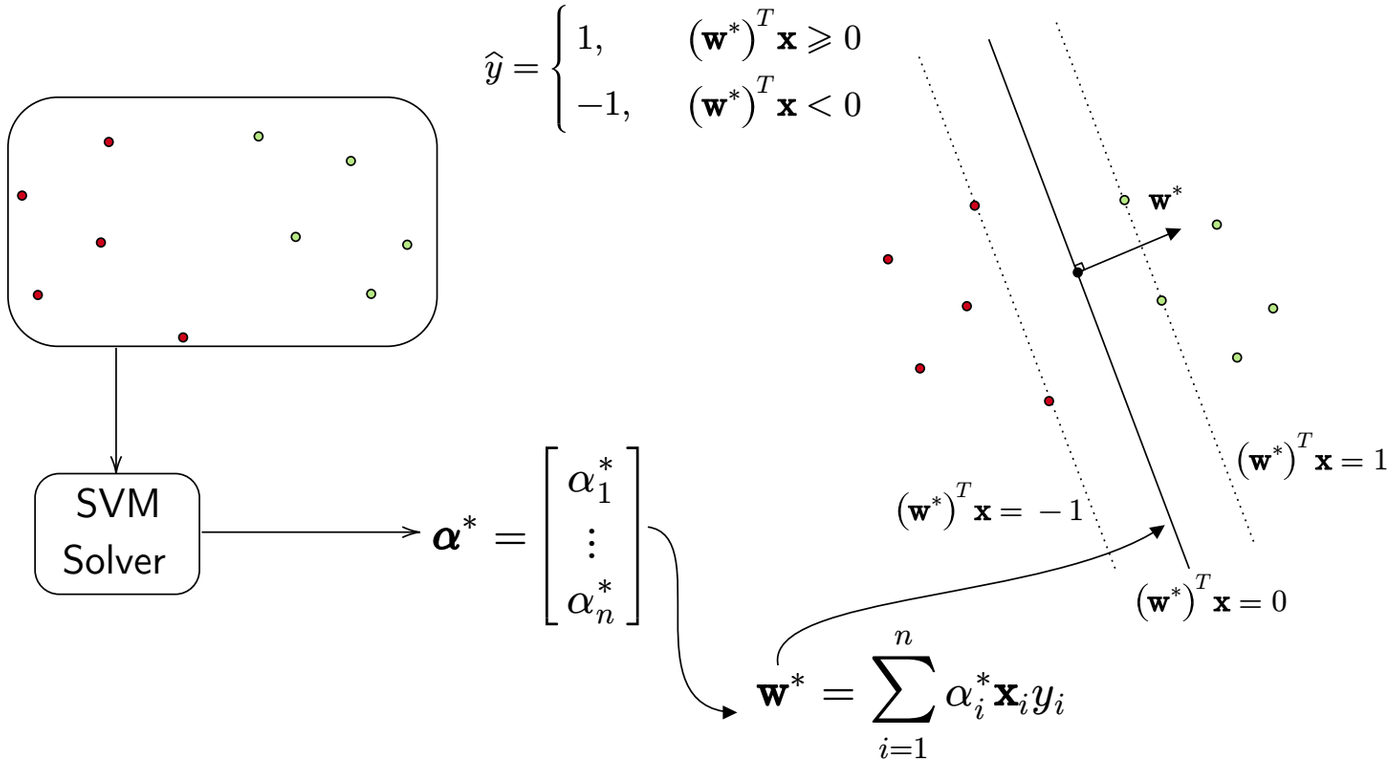
For a convex optimization problem, under certain conditions, the primal and dual have the same optimal value. Solving the dual problem is preferable for the following reasons:

- The constraints are simpler.
- The appearance of $\mathbf{X}^T\mathbf{X}$ points to kernels.

We don't discuss the solution method here. It is enough to know that a solution exists. If the optimal dual variable is denoted by $\boldsymbol{\alpha}^*$ and the optimal primal variable is denoted by $\mathbf{w}^*$, the two are related as:

$$\boxed{\mathbf{w}^* = \mathbf{X}\mathbf{y}\boldsymbol{\alpha}^*}$$

The schematic diagram that summarizes the discussion so far.



${\mathbf{w}^*}^T \mathbf{x} = 0$ is the decision boundary. ${\mathbf{w}^*}^T \mathbf{x} = \pm 1$ are called the supporting hyperplanes. We will see why this name is used in the next section.

3.4. Support Vectors

Complementary slackness conditions enforce the following equations:

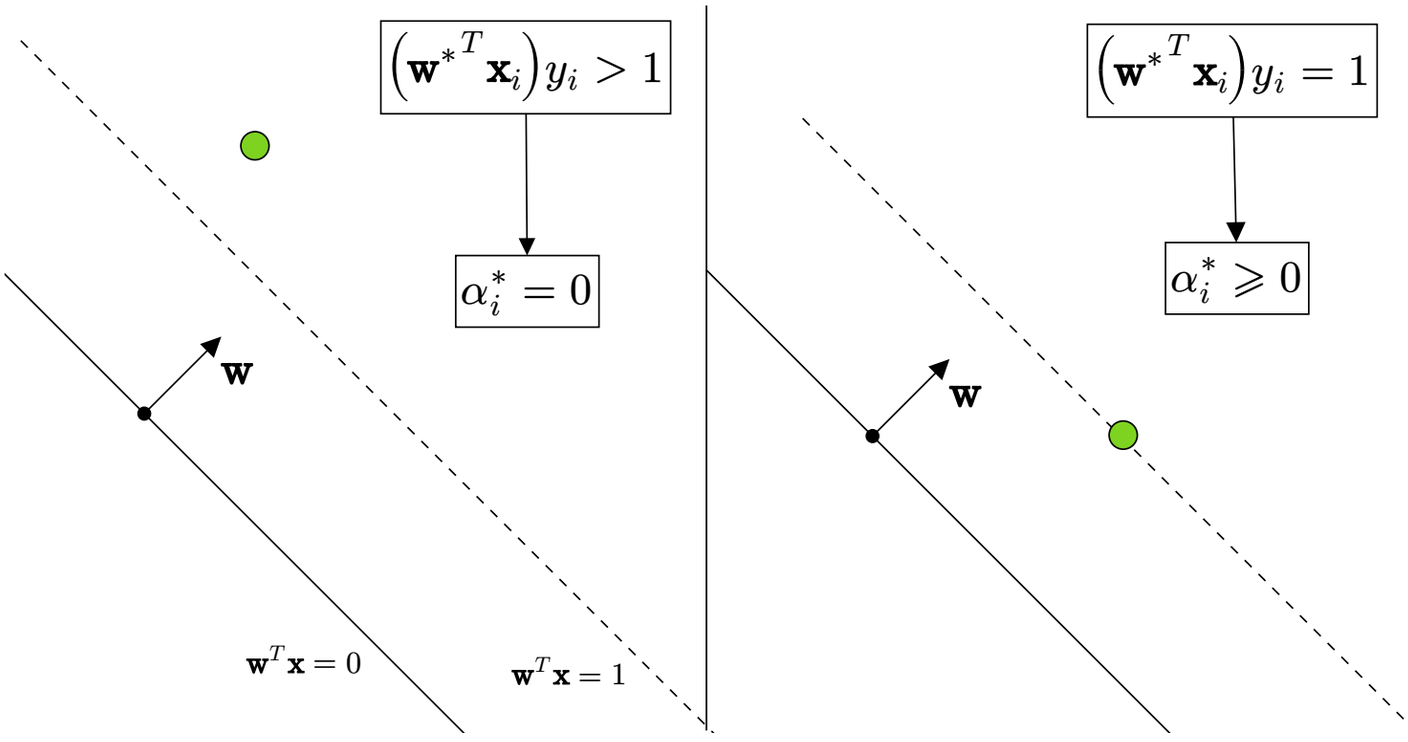
$$\alpha_i^* \left[1 - \left({\mathbf{w}^*}^T \mathbf{x}_i \right) y_i \right] = 0, \quad 1 \leqslant i \leqslant n$$

3.4.1. Primal View

- If $\left({\mathbf{w}^*}^T \mathbf{x}_i \right) y_i > 1$, we have $1 - \left({\mathbf{w}^*}^T \mathbf{x}_i \right) y_i < 0$, which forces $\alpha_i^* = 0$.

  For a point which lies beyond the supporting hyperplanes, the value of α^* is zero.

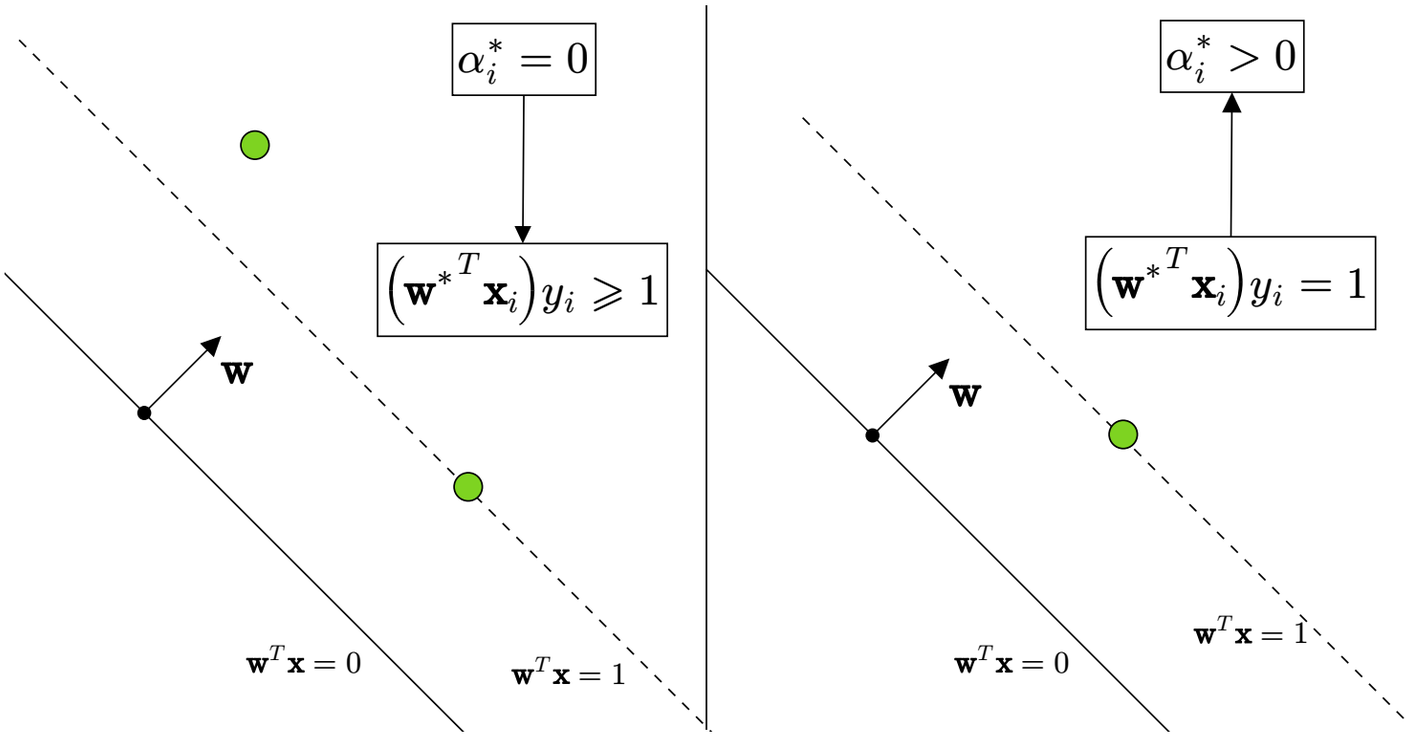
- If $\left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i = 1$, we have $1 - \left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i = 0$, which doesn't really force α_i^* to be a particular value. So we just settle for $\alpha_i^* \geqslant 0$. For a point which lies on the supporting hyperplanes, we can't comment on the value of α^*. It can be any non-negative quantity.



3.4.2. Dual view

- If $\alpha_i^* = 0$, we can't comment on the position of the point. It could either lie on the supporting hyperplane or beyond.

- If $\alpha_i^* > 0$, we have $1 - \left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i = 0 \implies \left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i = 1$. A point with $\alpha_i^* > 0$ is going to lie on one of the supporting hyperplanes.



Support Vectors
The points for which $\alpha_i^* > 0$ are called support vectors.

Some observations concerning support vectors:

- All support vectors lie on the supporting hyperplanes.
- But note that every point on the supporting hyperplanes need not be a support vector.
- The weight vector can now be seen as a sparse linear combination of the data-points. The sparsity comes from the fact that all non-support vectors have $\alpha_i^* = 0$.
- The support vectors are the most important training data-points as they lend their *support* in building the decision boundary.

4. Prediction

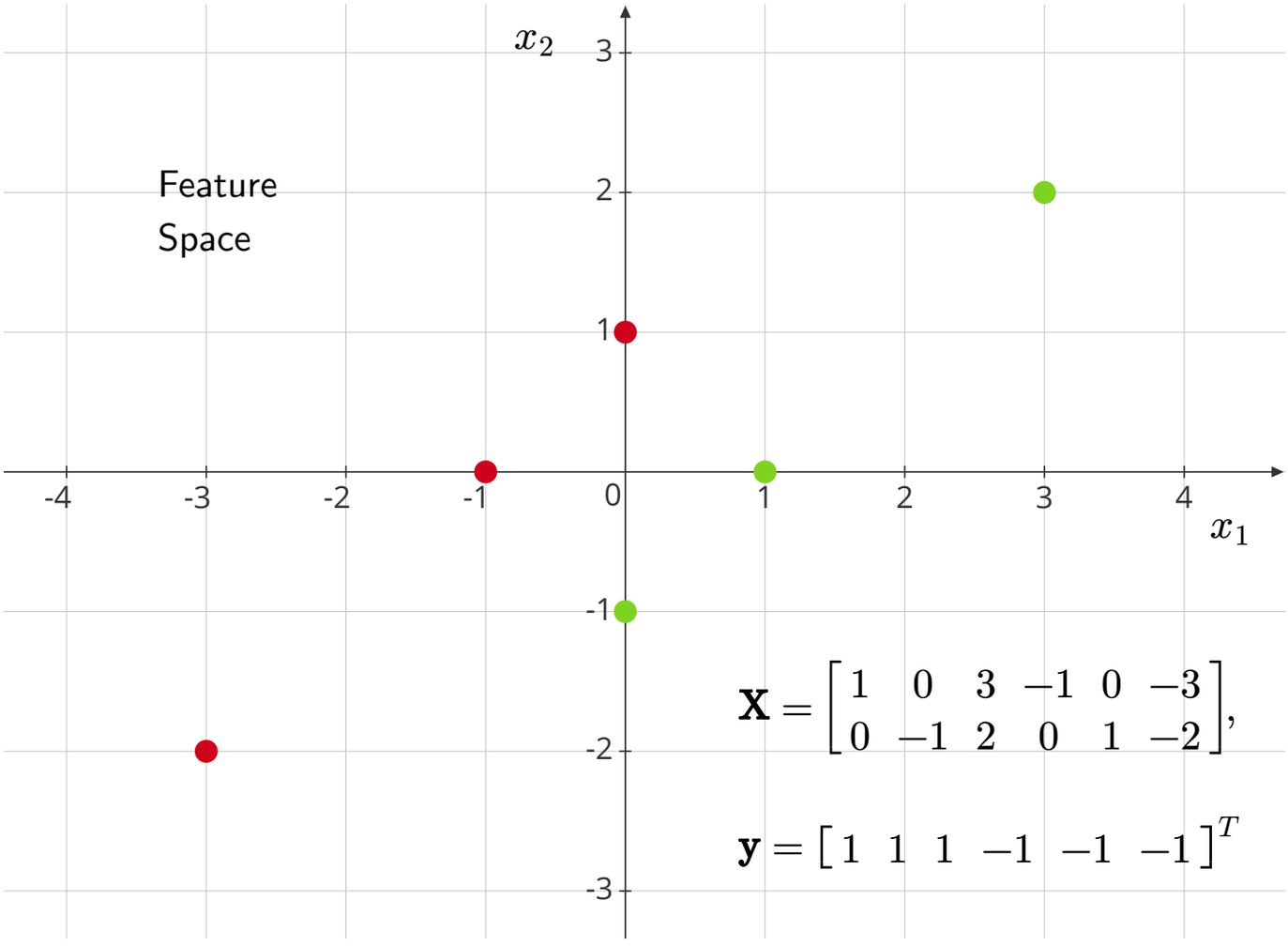
Once a SVM has been trained, it can be used to predict the label of a test point like any other linear classifier:

$$\widehat{y} = \begin{cases} 1, & \left(\mathbf{w}^*\right)^T \mathbf{x}_{\text{test}} \geqslant 0 \\ -1, & \left(\mathbf{w}^*\right)^T \mathbf{x}_{\text{test}} < 0 \end{cases}$$

Note that the constraint $\left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i \geqslant 1$ is binding on only the training data-points.

5. Example

Consider the following dataset:



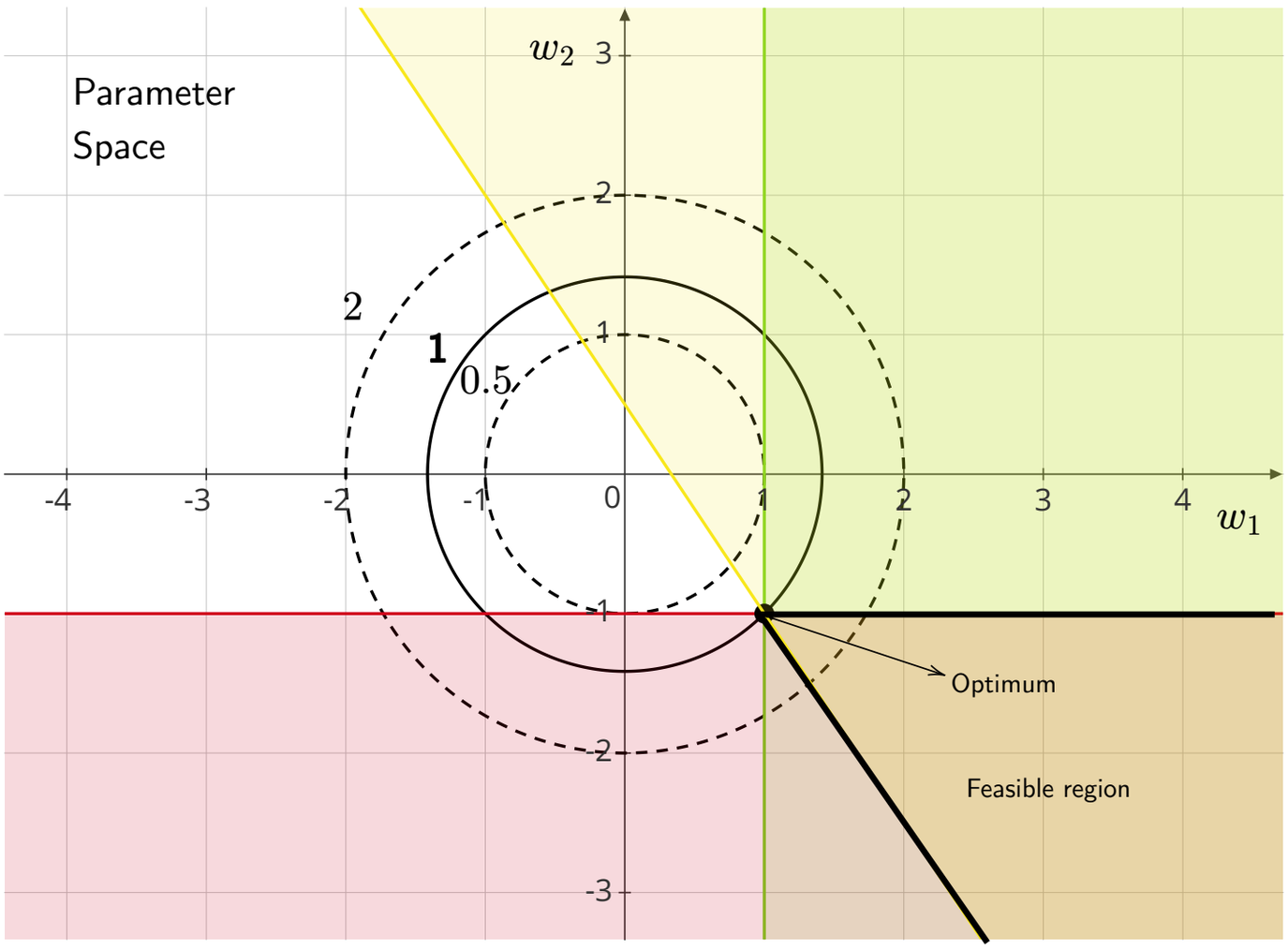
The optimization problem is:

$$\min_{\mathbf{w}} \quad \frac{w_1^2 + w_2^2}{2}$$

subject to:

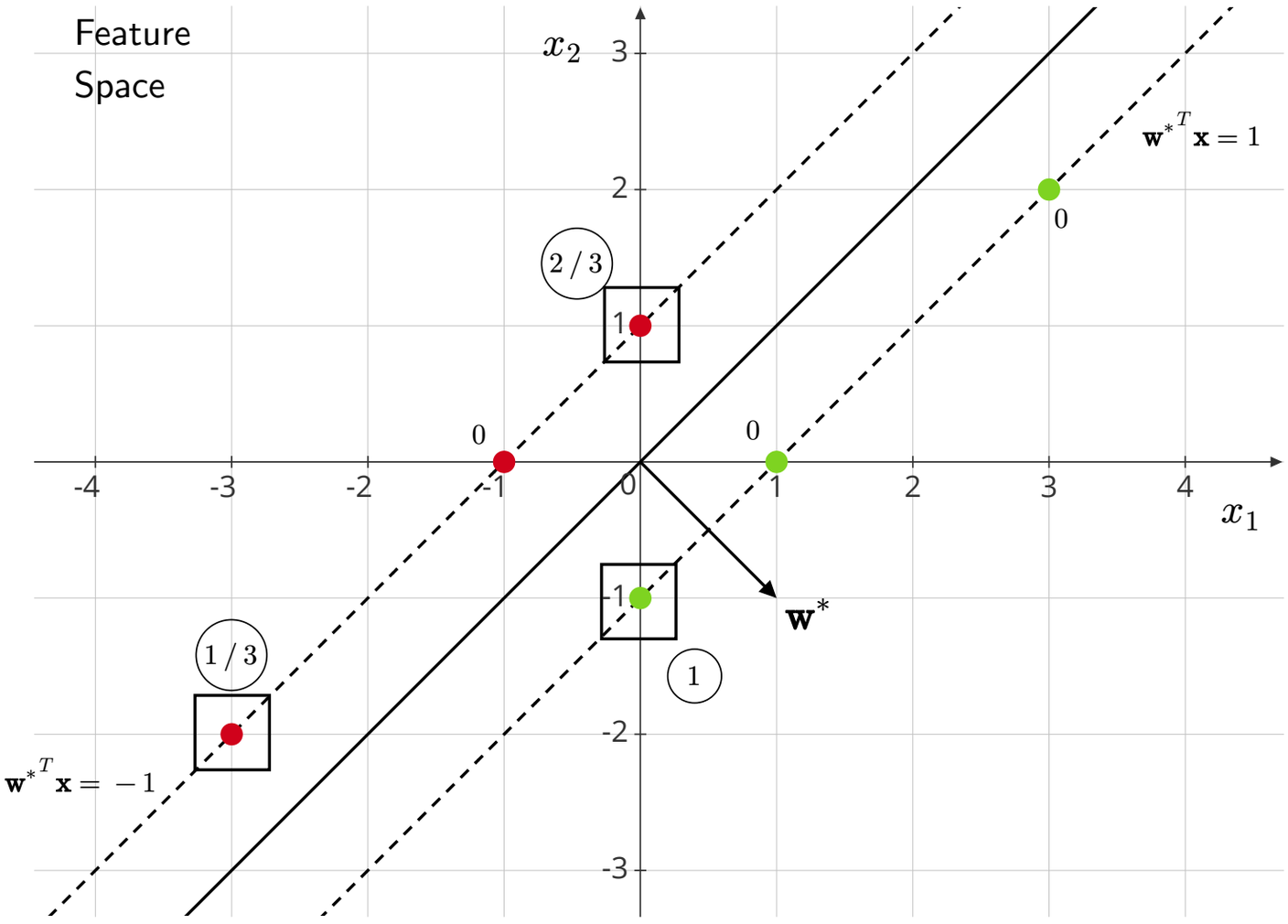
$$
\begin{aligned}
w_1 &\geqslant 1 && (1)\\
w_2 &\leqslant -1 && (2)\\
3w_1 + 2w_2 &\geqslant 1 && (3)\\
w_1 &\geqslant 1 && (4)\\
w_2 &\leqslant -1 && (5)\\
3w_1 + 2w_2 &\geqslant 1 && (6)
\end{aligned}
$$

On simplifying the constraints and plotting them along with the contours of the objective:



We see that the primal solution is $\mathbf{w}^* = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$. We can also solve for the dual.

One possible solution is $\boldsymbol{\alpha}^* = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 2/3 \\ 1/3 \end{bmatrix}$.



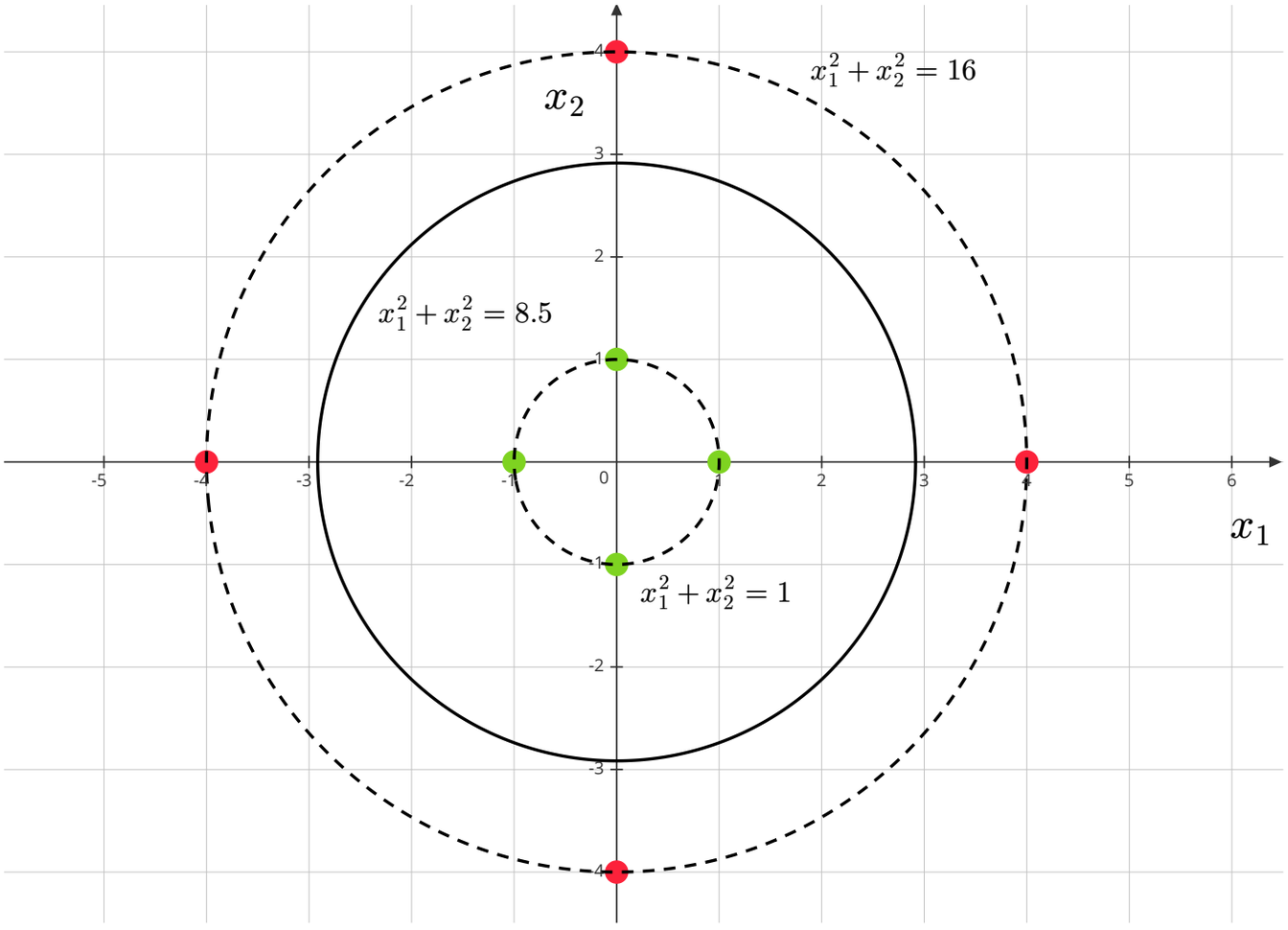
This is an example of a linear hard-margin SVM problem.

6. Kernel SVM

We have the following correspondence between linear SVM and kernel SVM. ϕ is typically a function that cannot be explicitly computed.

Linear SVM	**Kernel SVM**
$\mathbf{x}_i$	$\phi(\mathbf{x}_i)$
$\mathbf{X}$	$\phi(\mathbf{X})$
$\mathbf{X}^T\mathbf{X}$	$\mathbf{K}$
$\max\limits_{\boldsymbol{\alpha}\geqslant\mathbf{0}} \quad \boldsymbol{\alpha}^T\mathbf{1} - \frac{1}{2}\boldsymbol{\alpha}^T\left(\mathbf{Y}^T\mathbf{X}^T\mathbf{X}\mathbf{Y}\right)\boldsymbol{\alpha}$	$\max\limits_{\boldsymbol{\alpha}\geqslant\mathbf{0}} \quad \boldsymbol{\alpha}^T\mathbf{1} - \frac{1}{2}\boldsymbol{\alpha}^T\left(\mathbf{Y}^T\mathbf{K}\mathbf{Y}\right)\boldsymbol{\alpha}$
$\mathbf{w}^* = \mathbf{X}\mathbf{Y}\boldsymbol{\alpha}^*$	$\mathbf{w}^* = \phi(\mathbf{X})\mathbf{Y}\boldsymbol{\alpha}^*$
$\widehat{y} = \begin{cases} 1, & \left(\mathbf{w}^*\right)^T\mathbf{x}_{\text{test}} \geqslant 0 \\ -1, & \text{otherwise} \end{cases}$	$\widehat{y} = \begin{cases} 1, & \sum\limits_{i=1}^{n} \alpha_i^* k(\mathbf{x}_i, \mathbf{x}_{\text{test}}) \geqslant 0 \\ -1, & \text{otherwise} \end{cases}$

An example of a dataset that is not linearly separable in the original feature space but can be separated after a non-linear transformation. A quadratic kernel has been used here. The decision boundary and the two supporting hyperplanes (when brought back to the original feature space) are also displayed:



Note that the decision boundary is not equidistant to the two supporting curves. This is because the actual optimization happens in the transformed space ($\mathbb{R}^6$ in this example) and we are visualizing the result in the original space. In the transformed space, the corresponding supporting hyperplanes will indeed be equidistant to the boundary.